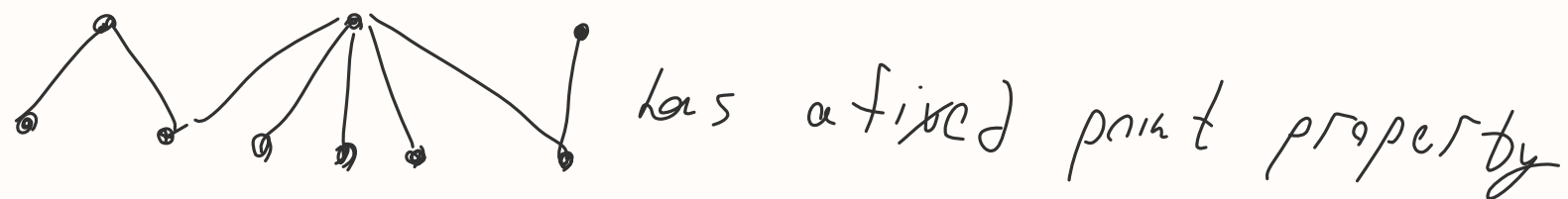


1-19 a

Prove that



has a fixed point property

Assume $\exists f$ - fixed point free endomorphism

$$e \leq b \Rightarrow f(e) \leq f(b) \Rightarrow \text{if } f(b) = e$$

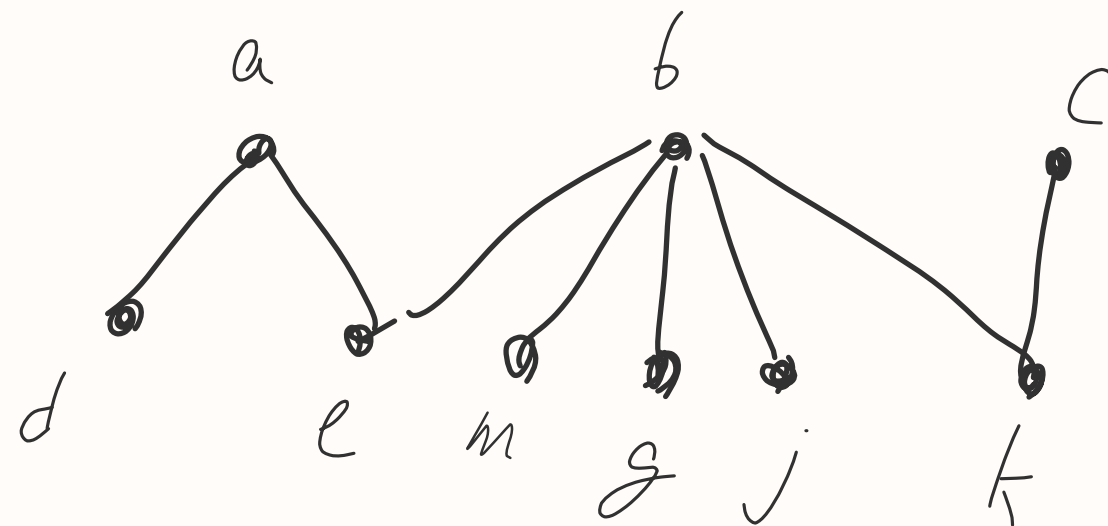
$$\text{then } f(e) = e \Rightarrow e \in \text{Fix}(f) \Rightarrow$$

$$f(b) \neq e, \text{ same way } f(b) \neq \begin{matrix} m \\ j \\ g \\ k \end{matrix}$$

$$f(b) \neq b \text{ or else } b \in \text{Fix}(f) \Rightarrow$$

$$f(b) \in \{d, a, c\}$$

Cases:



$$1) f(b) = d \Rightarrow f(e) = f(m) = f(j) = f(k) = d$$

$$a > e \Rightarrow f(a) = \begin{cases} d & \text{then } f(d) = d \Rightarrow d \in \text{Fix}(f) \\ a & \text{then } a \in \text{Fix}(f) \end{cases}$$

$$2) f(b) = a \Rightarrow f(e) \leq f(b) \Rightarrow f(e) = \begin{cases} a \\ e \\ d \end{cases} \text{ - contradicts } e \notin \text{Fix}(f)$$

but $a \not\leq e$

$$f(a) \geq f(e) \Rightarrow \begin{cases} \begin{cases} f(e) = a \\ f(a) = a \end{cases} - a \in \text{Fix} \\ \begin{cases} f(e) = d \\ f(a) = a \end{cases} - a \in \text{Fix} \\ \begin{cases} f(e) = d \\ f(a) = d \end{cases} \Rightarrow f(d) = d - d \in \text{Fix} \end{cases}$$

$$3) f(b) = c$$

$$f(k) \leq f(b) \Rightarrow f(k) = \begin{cases} c \Rightarrow f(c) = c - c \in \text{Fix}(f) \\ k - k \in \text{Fix} \end{cases}$$

"
 c